

IS 2207 – Electronics and Physical Computing

Credits – 3 (2L + 1P)

Course Description

This course introduces the principles and practices of electronics and physical computing for computing undergraduates. Students will learn the fundamentals of electronic circuits, digital logic, and signal processing, and explore how sensors, actuators, and microcontrollers enable interaction with the physical world. Emphasis is placed on hands-on experimentation with platforms such as Arduino and ESP32, alongside modern prototyping tools and IoT communication protocols. By the end of the course, students will be able to design, build, and test electronic systems that integrate hardware and software into real-world applications.

Course Evaluation: Assignments 40% + Paper 60%; Paper: 30 MCQs + 1 Structured

Learning Outcomes

By the end of this course, students will be able to:

- **Explain** fundamental concepts of voltage, current, resistance, power, and circuit laws.
- **Differentiate** between passive, active, and integrated components, and **analyze** their roles in circuits.
- **Apply** Boolean algebra, digital logic, and circuit principles to implement combinational and sequential systems.
- **Program** microcontrollers (Arduino, ESP32) to **control** sensors, actuators, and communication interfaces.
- **Integrate** wired and wireless communication protocols (UART, I²C, SPI, Wi-Fi, BLE, MQTT) into embedded applications.
- **Use** laboratory instruments and EDA/simulation tools to **test** and **evaluate** circuit performance.
- **Design** and **prototype** physical computing systems that combine hardware and software for real-world applications.
- **Assess** safety requirements and **demonstrate** best practices in electronics experimentation and prototyping.

Course Outline

1. Electronic Fundamentals

- Voltage, Current, Resistance, Power, Impedance
- Ohm's Law, KVL/KCL, Thevenin & Norton Theorems
- Signals: Analog/Digital, Frequency, Amplitude, Phase, Duty Cycle
- Modulation (ASK, FSK, PSK), Filters, Noise & Integrity
- DC & AC Circuits, RMS, Power Factor

2. Circuit Components

- **Passive:** Resistors, Capacitors, Inductors, Transformers, Relays, Fuses
- **Active:** Diodes, Transistors (BJT, MOSFET), Thyristors, Optoelectronics
- **Operational Amplifiers:** Modes & Applications
- **Power Systems:** Power supplies, Regulators, Batteries, DC-DC Converters

3. Digital Electronics

- Logic states, Gates, Families, ICs
- Boolean Algebra, Simplification, Karnaugh Maps
- Combinational Circuits (Adders, Multiplexers, Decoders)
- Sequential Circuits (Flip-flops, Counters, FSMs)
- Interfacing: A/D, D/A, Signal Conditioning
- Oscillators, Timers, Memory Devices

4. Sensors & Actuators

- Sensors: Temperature, Light, Motion, Pressure, Distance, Biosensors
- Actuators: Motors, Solenoids, Buzzers, Displays (LCD, OLED)
- Interfacing considerations & calibration

5. Physical Computing

- Concepts: Embedded Systems, IoT, Edge Computing
- Microcontrollers: Architecture & Families (ARM, AVR, ESP32, PIC)
- Programmable Logic: FPGA basics, VHDL/Verilog
- Arduino/ESP32: I/O, PWM, Sensor/Actuator Interfacing, Communication
- Protocols: UART, SPI, I2C, USB, Wi-Fi, BLE, ZigBee, MQTT, LoRa
- Low-Power Design: Sleep Modes, Energy Harvesting
- Advanced Platforms: Raspberry Pi, Jetson Nano, Cloud IoT

6. Practical Electronics & Prototyping

- Safety & Best Practices
- Measurement Tools: Multimeter, Oscilloscope, Function Generator, PSU
- Prototyping Tools: Breadboards, Soldering, PCB Design
- EDA & Simulation: LTspice, KiCad, Proteus, Fritzing
- Datasheets & Electronic Catalogs for Design

References

- Scherz, P., & Monk, S.: ***Practical electronics for inventors***. McGraw-Hill Education.
- Monk, S.: ***Hacking electronics: Learning electronics with Arduino and Raspberry Pi***. McGraw-Hill Education.
- ***Electronics tutorials***. <https://www.electronics-tutorials.ws/>